

Production Supporting Systems in Factories

ระบบสนับสนุนการผลิตในโรงงานอุตสาหกรรม

Self-Hosting Node-RED

Advantages

- Greater performance
 - Fast to deploy
- Custom nodes
- Development environment
 - Test flows locally before deploying to production

Disadvantages

- Need to configure local tunnel setup to expose local Node-RED server to the internet. (ERPNext -> Node-RED connection)

Setting up Node-Red

Software

- Install [Node.js](#) (Version 24)
- Install [Visual Studio Code](#)

VSCode

- Change default terminal in `VSCode` (Windows only)
 - Open a window
 - Select `terminal` -> `New Terminal`
 - press `ctrl` + `p`
 - Type `> Terminal: Select Default Profile` and click
 - Select `cmd`

Package Manager

- `Node.js` comes with `npm` package manager.
 - But it is quite slow.
- Let's use a more performant package manager call `pnpm`.
 - `npm install -g pnpm`

Installing Node-Red

- Create a new folder
- Drag a folder into VSCode
- Select terminal -> New Terminal

Installing Node-Red

- `pnpm init`
- `pnpm install node-red @flowfuse/node-red-dashboard node-red-contrib-telegrambot node-red-contrib-influxdb`
- `pnpm approve-builds`

Modifying `package.json`

```
{  
  "scripts": {  
    "start": "node-red -u ./local -p 1880"  
  }  
}
```

Starting Node-Red

- `pnpm run start`
- Visit the web browser at
 - `http://127.0.0.1:1880` or
 - `http://localhost:1880`

Advanced Setup

| Using `nvm` and `Chocolatey` on Windows

Step 1/5

- Open PowerShell as Administrator
 - Press the Windows + R
 - Type powershell .
 - Press Ctrl + Shift + Enter .
- Change Execution Policy to allow running scripts
 - Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope LocalMachine

Step 2/5

- Install `Chocolatey` package manager
 - `Set-ExecutionPolicy Bypass -Scope Process -Force;`
`[System.Net.ServicePointManager]::SecurityProtocol =`
`[System.Net.ServicePointManager]::SecurityProtocol -bor 3072; iex ((New-`
`Object`
`System.Net.WebClient).DownloadString('https://community.chocolatey.org/ins`
`tall.ps1'))`

Step 3/5

- Install `nvm` and `vscode`
 - `choco install nvm vscode`
- Close and reopen `PowerShell` as normal user
 - `nvm install 24; nvm use 24; npm install -g pnpm`

Step 4/5

- Create project folder on Desktop

- `cd "$env:USERPROFILE\Desktop"; New-Item -ItemType Directory -Force -Path node-red; cd node-red; code .`

- In VSCode terminal

- `pnpm init; pnpm install --dangerously-allow-all-builds node-red @flowfuse/node-red-dashboard node-red-contrib-telegrambot node-red-contrib-influxdb`

Step 5/5

- Modify `package.json` to add start script

```
$json = Get-Content package.json -Raw | ConvertFrom-Json;  
$json.scripts | Add-Member -NotePropertyName "start" -NotePropertyValue "node-red -u ./local -p 1880"  
$json | ConvertTo-Json -Depth 10 | Set-Content package.json
```

- Start `Node-RED`
 - `pnpm run start`